

Ali Abdallah

Junior Software Engineer

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🛡 Military Service: **Completed**

Computer Science graduate with strong foundations in OOP, Data Structures, Algorithms, Design Patterns, and Databases. Passionate about Data Science, NLP, and Machine & Deep Learning.

Education

B.Sc. in Computer Science, Ain Shams University in *Egypt* (Class of 2024). October 2020 - July 2024
Grade: Very Good
Graduation Project: Masha'erohom - Arabic Emotion Detector (Grade: A+)

Skills

Programming & Tools:	Core CS Concepts:	Data Science & AI:
● Python ● R ● C++ ● C# ● Java ● SQL, SQL Lite ● MongoDB ● Git ● Bash	● OOP ● Data Structures ● Algorithms ● Design Patterns ● System Design Principles ● Operating Systems	● Data Analysis ● Statistical Analysis ● Data Visualization ● Data Mining ● Machine Learning ● Deep Learning ● NLP ● Data Security

Experience & Projects

Masha'erohom - Arabic Emotion Detector  August 2023 - July 2024

- Led a team to develop an Arabic emotion detection model exceeding state-of-the-art (SOTA) accuracy (92%, +7% vs. prior work).
- Developed and architected an ensemble of Bi-LSTM, Bi-GRU, and fine-tuned MARBERTv2, stacked via Random Forest after extensive hyperparameter experimentation.
- Fine-tuned pretrained Arabic word embeddings to better capture dialectal nuances, diacritics, and morphology.
- Built a Flask web app that ingests Twitter data and visualizes public emotion distributions for user-entered keywords.
- Published a **peer-reviewed paper**  following formal evaluations in IJICIS.

ElectroHub – Online Electronics Shopping System  November 2023 - December 2023

A comprehensive e-commerce system developed with Java, JSP, and MySQL, applying design patterns (Singleton, Factory, Strategy).

English Sign Language Recognition  August 2023 - September 2023

Using landmark detection and CNNs, built models achieving 99% test accuracy. Deployed the final model using Random Forest for real-time predictions.

Data Security Algorithms  February 2023 - May 2023

Implementation of 15 algorithms in C#, including AES, DES, Playfair, and others. We conducted tests to validate their functionality.

Smoking Detection

April 2023 - May 2023

The Data Analytics project in it we perform data analysis on bio-signals dataset to detect and monitor smoking behavior using machine learning classification models.

Spam Detection

April 2023 - April 2023

A website  that you can predict through if a message is spam or not.

Loan Prediction

February 2023 - February 2023

This project aims to predict if the loan will be accepted or not. Implemented by Jupyter notebook in Python.

FOS

October 2022 - January 2023

FOS is a simple operating system implemented in C.

Customer Segmentation

November 2022 - December 2022

Implemented a customer segmentation analysis project using R, leveraging techniques such as k-means, DBSCAN, and hierarchical clustering. Cleaned and preprocessed data conducted exploratory data analysis, and visualized results through various plots and charts. Developed valuable insights into business strategy during the Statistical Inference course at the college.

Mini Wasalni

April 2022 - June 2022

C++ program that allows the user to enter the graph "map" which represents the towns then find him the shortest path to reach his destination from any given source town.